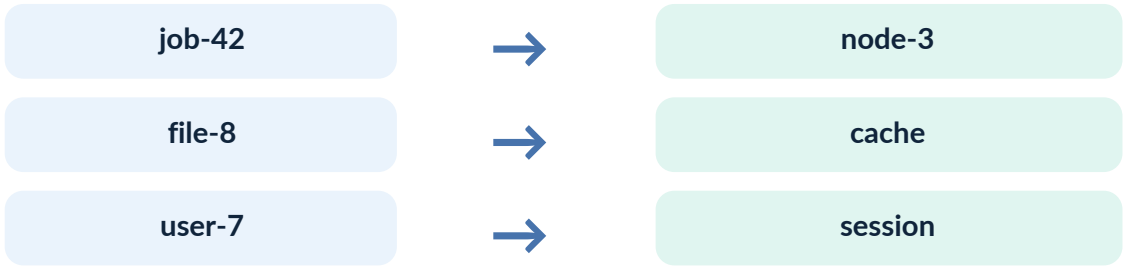
HASH MAPS, CACHES & SETS

Avoid searching every computer or recalculating the same answer.

SCAN / LEARN

THE SYSTEM AT A GLANCE



FOUR IDEAS THAT MATTER

1 HASH MAP (KEY → VALUE)

Store an association such as job ID → assigned server. A hash function helps locate the key efficiently.

2 CACHE (FAST COPY)

Keep a recent or popular value near where it is needed. The cache may become outdated.

3 SET (UNIQUE MEMBERS)

Keep unique identifiers: servers already visited or active connections. It prevents duplicate entries.

4 CACHE INVALIDATION

A cached answer is useful only while it is valid for its context. Changes, time and permissions matter.

MEMORY HOOK

A dictionary answers “Where is job 42?”; a cache asks “Can I reuse it safely?”

DISCUSS WITHOUT COMPUTERS

Why might yesterday’s cached inventory give the wrong answer today?

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